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EVALUATION

I. Choose the best answer

1. The amount of heat exchanged with the surrounding at constant temperature and pressure is given by the quantity

   a) ΔE b) ΔH c) ΔS d) ΔG

2. All the naturally occurring processes proceed spontaneously in a direction which leads to

   a) decrease in entropy b) increase in enthalpy

   c) increase in free energy d) decrease in free energy

3. In an adiabatic process, which of the following is true ?

   a) $q = w$ b) $q = 0$ c) $\Delta E = q$ d) $P \Delta V = 0$

4. In a reversible process, the change in entropy of the universe is

   a) > 0 b) ≥ 0 c) < 0 d) $= 0$

5. In an adiabatic expansion of an ideal gas

   a) $w = -\Delta u$ b) $w = \Delta u + \Delta H$ c) $\Delta u = 0$ d) $w = 0$

6. The intensive property among the quantities below is

   a) mass b) volume c) enthalpy d) $\frac{\text{mass}}{\text{volume}}$

7. An ideal gas expands from the volume of 1×10^{-3} m^3 to 1×10^{-2} m^3 at 300 K against a constant pressure at 1×10^5 Nm^{-2}. The work done is

   a) – 900 J b) 900 kJ c) 270 kJ d) – 900 kJ

8. Heat of combustion is always

   a) positive b) negative c) zero d) either positive or negative

9. The heat of formation of CO and CO_2 are – 26.4 kCal and – 94 kCal, respectively. Heat of combustion of carbon monoxide will be

   a) + 26.4 kcal b) – 67.6 kcal c) – 120.6 kcal d) + 52.8 kcal